Joshua Dinn

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Education

Monash University ,*Melbourne, Australia* Mar 2023 – Oct 2027
Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)
WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

Experience

Collingwood Football Club ,*Data & Analytics Intern* Mar 2026 – Jun 2026

- Built a membership churn prediction model in XGBoost and engineered the feature pipelines from historical transaction and engagement data.
- Built an interactive dashboard visualising churn drivers and SHAP feature importance for commercial stakeholders.
- Designed and implemented a repository synchronisation workflow between the club's web and mobile codebases to streamline deployments.

Monash eSolutions ,*AI Fitness Trainer* Jul 2026 – Present

- Run in-person sessions teaching students and staff how to use AI effectively, ethically, and in everyday study and work.
- Assist in delivering a university webinar on AI to an audience of 200+ attendees.

Monash Deep Neuron ,*Optimised Computing Member* Sep 2025 – Present

- Building the on-device AI memory system for AR glasses: a vector database with HNSW indexing and approximate nearest-neighbour (ANN) search powering a retrieval-augmented generation (RAG) pipeline that grounds responses in prior context with low latency.
- Teaching Assistant for an 8-week C++ and Accelerated Computing workshop on GPU programming, CUDA, and HPC, run with NVIDIA, Pawsey Supercomputing Centre, and NCI.
- NVIDIA CUDA certified.

Outlier AI ,*AI Trainer* Jun 2024 – Dec 2025

- Reviewed and refined AI-generated solutions written in LaTeX for logical correctness and technical clarity.
- Gave structured feedback on reasoning and formatting to improve model output quality.

Projects

StdOut: Real-Time AI Technical Interview Simulator Feb 2026

- Built a real-time interview simulation platform with a 3-person hackathon team using React, Node.js, MongoDB, and WebSockets.
- Engineered a low-latency pipeline combining live transcription, AI-driven conversation, and in-browser coding, handling streaming updates asynchronously so they never blocked the session.
- Built a Python code-execution terminal with event-based tracking of code diffs and session timelines.

Interactive Maze Solver & Algorithm Visualiser Oct 2025

- Built a pathfinding platform visualising 7 algorithms across procedurally generated mazes.
- Added automated benchmarking and metrics tracking across 125,000+ simulated maze runs.

Algorithmic Trading System for S&P 500 ETF May 2025

- Built and backtested a trading system using Hidden Markov Models (HMMs) to classify market regimes for SPY.
- Integrated RSI, ATR, and volatility signals into position sizing and trade logic.
- Deployed on an AWS Linux server with the Alpaca API for automated paper trading.

Skills & Interests

Languages: Python, C++, JavaScript, TypeScript, Java, MATLAB, SQL
Frameworks & Tools: React, Node.js, MongoDB, AWS, Linux, Git, pandas, NumPy
Interests: Distributed systems, machine learning, backend engineering, statistical modelling